

PYTHON

Getting Started Guide

Mastering the Setup and Execution

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Setting Up Your Environment

Before diving into code, you need to ensure Python is accessible on your system. Python is cross-platform, meaning it works on Windows, macOS, and various Linux distributions.

Checking Installation:

To see if you already have Python installed, open your terminal (Command Prompt on Windows, Terminal on Mac/Linux) and type:

- Windows: ``python --version`` or ``py --version``
- Mac/Linux: ``python3 --version``

If a version number (e.g., Python 3.10.x) appears, you are ready to go. If not, visit the official website at python.org to download the latest installer.

Your First Script: 'Hello World'

Python development typically follows a simple workflow: Write code in a text editor, save it with a `.py` extension, and run it through the interpreter.

Step 1: Open any text editor (Notepad, TextEdit, or VS Code).

Step 2: Type exactly: ``print("Hello, World!")``

Step 3: Save the file as ``hello.py``.

Step 4: Navigate to the folder in your terminal and run: ``python hello.py``.

Outcome: The terminal will display 'Hello, World!'. This marks your successful entry into the world of programming!

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The Interactive Shell (REPL)

Sometimes, you want to test a single line of code without creating a file. Python offers the 'Read-Eval-Print Loop' (REPL), also known as the Python Command Line.

To enter the shell, simply type ``python`` (or ``py`` / ``python3``) into your terminal. You will see the ``>>>`` prompt, indicating Python is waiting for your command.

Examples of REPL usage:

- Simple Math: ``>>> 5 + 5`` (Result: 10)
- Quick Tests: ``>>> print('Testing...')`` (Result: Testing...)

To exit the REPL, type ``exit()`` or use ``Ctrl+Z`` (Windows) / ``Ctrl+D`` (Mac/Linux).

Checking Versions via Code

In professional development, you might need to ensure your script is running on a specific version of Python. You can check this programmatically using the built-in ``sys`` module.

Example Code:

```
```python
import sys

print(sys.version)
```
```

This script will output the exact release version and compiler information used to run the current process.

Choosing the Right Editor

While basic text editors work, 'Integrated Development Environments' (IDEs) make coding easier with features like syntax highlighting and error detection.

Popular Choices for CodeHive Students:

- VS Code: Lightweight and powerful with the Python extension.
- PyCharm: A professional IDE specifically designed for Python.
- Jupyter Notebooks: Excellent for data science and step-by-step experimentation.
- Thonny: A simple IDE perfect for absolute beginners.

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Common Troubleshooting

If the 'python' command is not recognized on Windows, ensure you check the box that says 'Add Python to PATH' during the installation process.

On macOS, Python 2 was often pre-installed. Always use the ``python3`` command to ensure you are using the modern version of the language.

Case Sensitivity: Remember that Python is case-sensitive. ``Print()`` will cause an error, while ``print()`` will work perfectly.

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The CodeHive Path Forward

Now that you have Python installed and running, you've crossed the biggest hurdle for most beginners.

Next Steps:

- Learn about Variables and Data Types.
- Master Control Flow (If statements and Loops).
- Explore Python Functions and Modules.

The journey from a 'Hello World' script to building complex AI or Web Apps starts with these fundamentals.

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